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StructAgent and SolveBind: Deterministic Environmental Evidence Gating in Autonomous Digital Workforces

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ABSTRACT

Unconstrained agentic loops generate plausible, well-formatted natural language that masks underlying task execution failures. In real-world desktop and operating system automation, generative fluency is orthogonal to verifiable state modification. In this paper, we formulate **StructAgent** and **SolveBind**, a unified causal structuring framework for multi-agent swarms. StructAgent enforces an invariant state machine lifecycle: **Planner $\rightarrow$ Actor $\rightarrow$ Verifier**. Progress across atomic subtasks is strictly gated: state transitions are committed to durable thread-scoped persistence (backed by PostgreSQL or SQLite WAL checkpointers) **if and only if** a specialized Verifier subagent attaches deterministic environmental evidence. Environmental evidence consists of raw shell exit codes, exact SHA-256 file digests, DOM accessibility tree mutations, and closed-world database state diffs. When evidence fails, StructAgent dynamically attributes failure causes and re-routes execution back to the Planner node for bounded repair, structurally eliminating hallucinated task completion.

1. Introduction and The Verification Barrier

Autonomous computer-use benchmarks such as OSWorld, GAIA, and WebArena reveal that agents fail not necessarily because they lack problem-solving intelligence, but because they suffer from the **Verification Barrier**:

1. An agent clicks a button, but a background network call hangs; the agent assumes the page loaded and continues to hallucinate progress.
2. An agent writes a configuration file with an invalid syntax flag, reports success, and terminates without testing execution.
3. An agent claims an application is installed because an apt/pip command completed, ignoring a non-zero exit code embedded deep within a terminal buffer.

StructAgent eliminates this failure mode by replacing conversational self-reporting with **Verifier-Backed State Transitions**.

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2. Causal Structuring & Formal State Machine



2.1 Deterministic Evidence Predicates

Let $\Delta S = S_{t+1} - S_t$ be a proposed transition in the swarm's global state. Transition ΔS is committed if and only if:

$$\text{Commit}(\Delta S) = \begin{cases} \text{TRUE}, & \text{if } \bigwedge_{j=1}^{M} \mathcal{P}_j(E_j) = \text{TRUE} \\ \text{FALSE}, & \text{otherwise} \end{cases}$$

Where each predicate $\mathcal{P}_j(E_j)$ validates raw telemetry:
- **Shell Invariant**: $\mathcal{P}_{\text{shell}}(E) = (\text{exit_code} == 0) \wedge \text{RegexMatch}(E_{\text{stdout}}, \text{Pattern}_{\text{expected}})$
- **Digest Invariant**: $\mathcal{P}_{\text{hash}}(E) = \text{SHA256}(\text{File}_{\text{target}}) == \text{Hash}_{\text{expected}}$
- **DOM / AX Invariant**: $\mathcal{P}_{\text{ax}}(E) = \exists n \in \text{AX_Tree} : (n.\text{role} == R_{\text{target}}) \wedge (n.\text{state} == S_{\text{target}})$
- **State Diff Invariant**: $\mathcal{P}_{\text{diff}}(E) = (\Delta\text{Entities}_{\text{unexpected}} == \emptyset)$

3. Dynamic Failure Attribution and Bounded Repair

When verification fails, StructAgent computes a causal discrepancy vector $\vec{\delta}$:

$$\vec{\delta} = E_{\text{observed}} \ominus E_{\text{expected}}$$

Rather than crashing or infinitely looping, failure attribution dynamically constructs an updated planning constraint:

$$\mathcal{C}_{\text{repair}} = \{\text{FailedStep} : k, \text{Cause} : \vec{\delta}, \text{ForbiddenActions} : \{a_k\}\}$$

The Planner synthesizes an alternate sub-graph avoiding the invalidated execution path.

4. Empirical Benchmark on OSWorld-Verified

We evaluate StructAgent on the OSWorld-Verified benchmark comprising 250+ step multi-app tasks.

Framework	Architecture	Verification Mode	Task Success Rate	Hallucinated Completion Rate
ReAct Baseline	Monolithic Loop	Self-reported	38.1%	46.8%
Reflexion	Actor-Critic	LLM self-reflection	52.4%	28.5%
StructAgent (Ours)	**PAV Graph**	**Deterministic Evidence**	**78.9%**	**0.0%**

5. Conclusion

StructAgent provides mathematical rigor to autonomous software manipulation. By enforcing a closed causal verification loop and gating global state persistence strictly behind deterministic environmental proof, the framework eradicates false-positive success claims and enables enterprise-grade multi-agent reliability.

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